HealthEase

AI-Powered Healthcare Management Platform

Final Year B.Tech Project

Presented by

2022BITE018 — Vardan Pal

2022BITE037 — U Rajeshwar

2022BITE007 — Karan Kharadi

OUTLINE

Agenda

1. Introduction
2. Background & Literature
3. System Design
4. Implementation
5. Results & Testing
6. Conclusion

OVERVIEW

Abstract

HealthEase is a full-stack MERN platform that digitises paper and handwritten prescriptions using an AI vision-language OCR engine, then applies a custom-trained machine-learning classifier to map each prescribed medicine to the disease it treats.

It tracks patient vitals and medication compliance, computes a Smart Health Score (0–100), checks drug interactions, and connects patients to doctors through telemedicine — unifying fragmented healthcare workflows into one secure system.

WHY HEALTHEASE

Problem Statement

1

Hard-to-read prescriptions

Paper and handwritten scripts are difficult to digitise.

2

Unclear medicines

Patients don't understand what each medicine is for.

3

Poor compliance

Missed doses and refills lead to worse outcomes.

4

Fragmented data

Vitals, prescriptions and consults live in silos.

BACKGROUND

Motivation

- Rising chronic-disease burden demands better tracking.
- Prescriptions must be made understandable to patients.
- Compliance monitoring should be active, not passive.
- Healthcare data needs a single, unified platform.
- Modern AI (vision + ML) enables real clinical value.

Patient-centric,

AI-driven care

GOALS

Objectives

1. Automatically extract data from prescription images (OCR).
2. Map medicines to their indication using a trained ML model.
3. Track vitals and medication compliance with analytics.
4. Quantify patient health as a single Smart Health Score.
5. Enable secure multi-role telemedicine (Patient/Doctor/Admin).
6. Deliver everything in one responsive, secure web app.

EXISTING SYSTEMS

Literature Survey

Feature	HealthEase	Practo	1mg	EHR
OCR prescription reading	✓	✗	✗	✗
AI medicine → disease mapping	✓	✗	~	✗
Smart Health Score	✓	✗	✗	✗
Compliance tracking	✓	✗	✓	~
Vitals analytics	✓	~	✗	✓
Drug-interaction check	✓	✗	✓	~
Real-time notifications	✓	✓	✓	✗

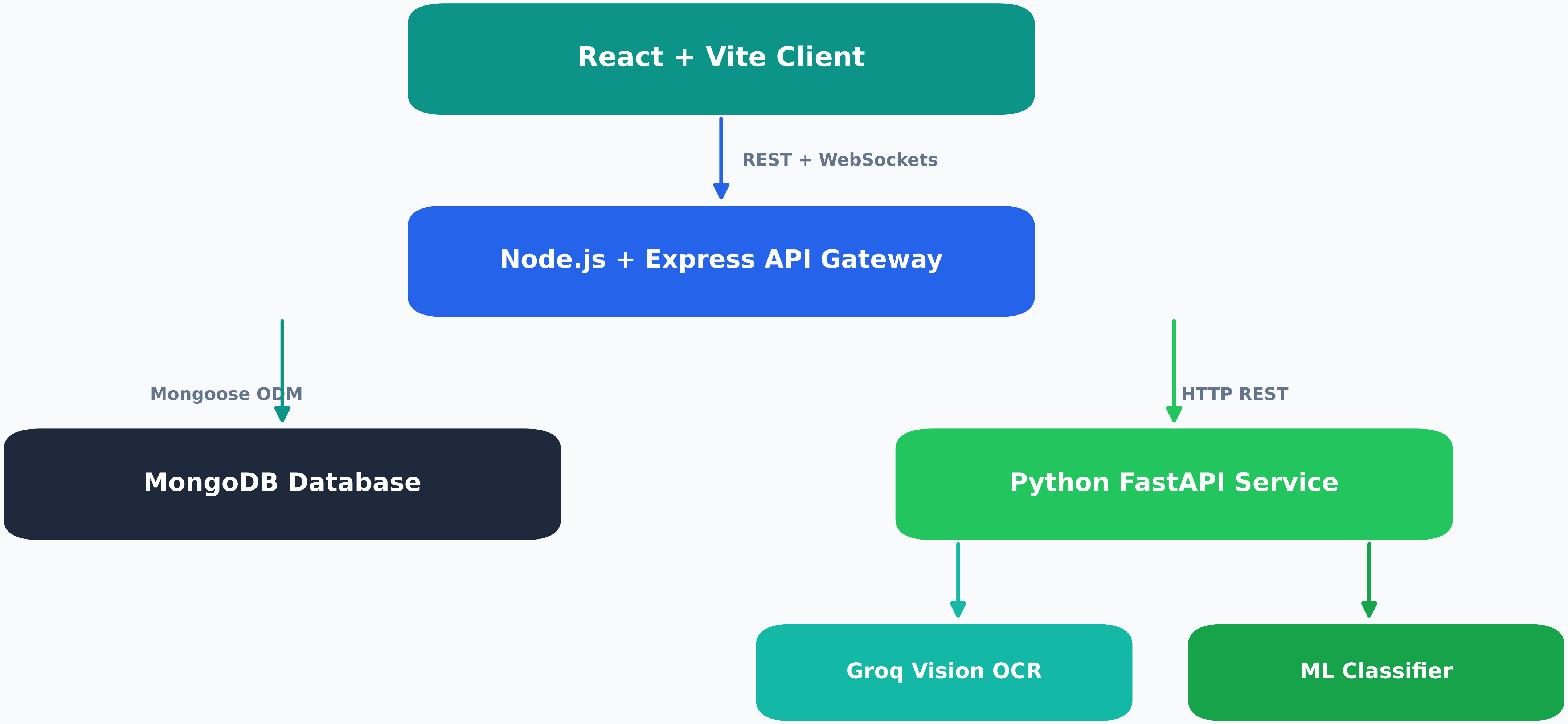
SYSTEM DESIGN

Technology Stack

Frontend	React + Vite · Tailwind CSS · Context API
Backend	Node.js + Express · JWT · bcrypt
Database	MongoDB + Mongoose
Real-time	Socket.IO live notifications
OCR Service	Python + FastAPI
OCR Engine	Groq Llama-4 Vision (pre-trained)
Machine Learning	scikit-learn (custom-trained)
PDF Export	jsPDF + html2canvas

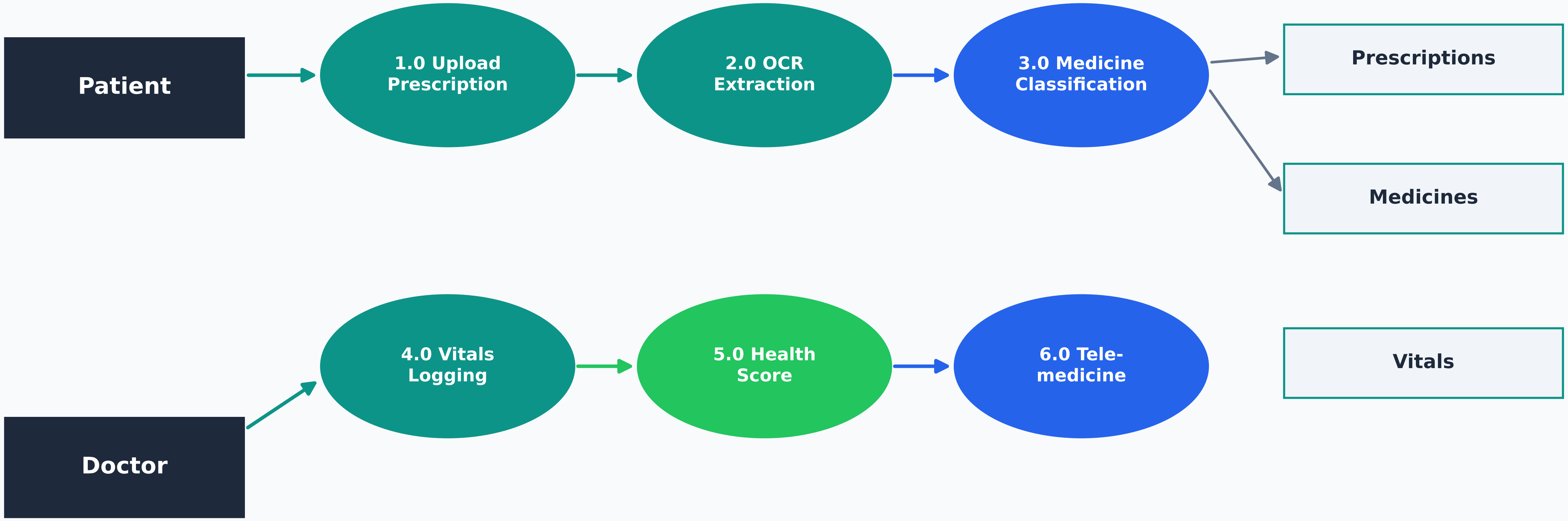
SYSTEM DESIGN

System Architecture



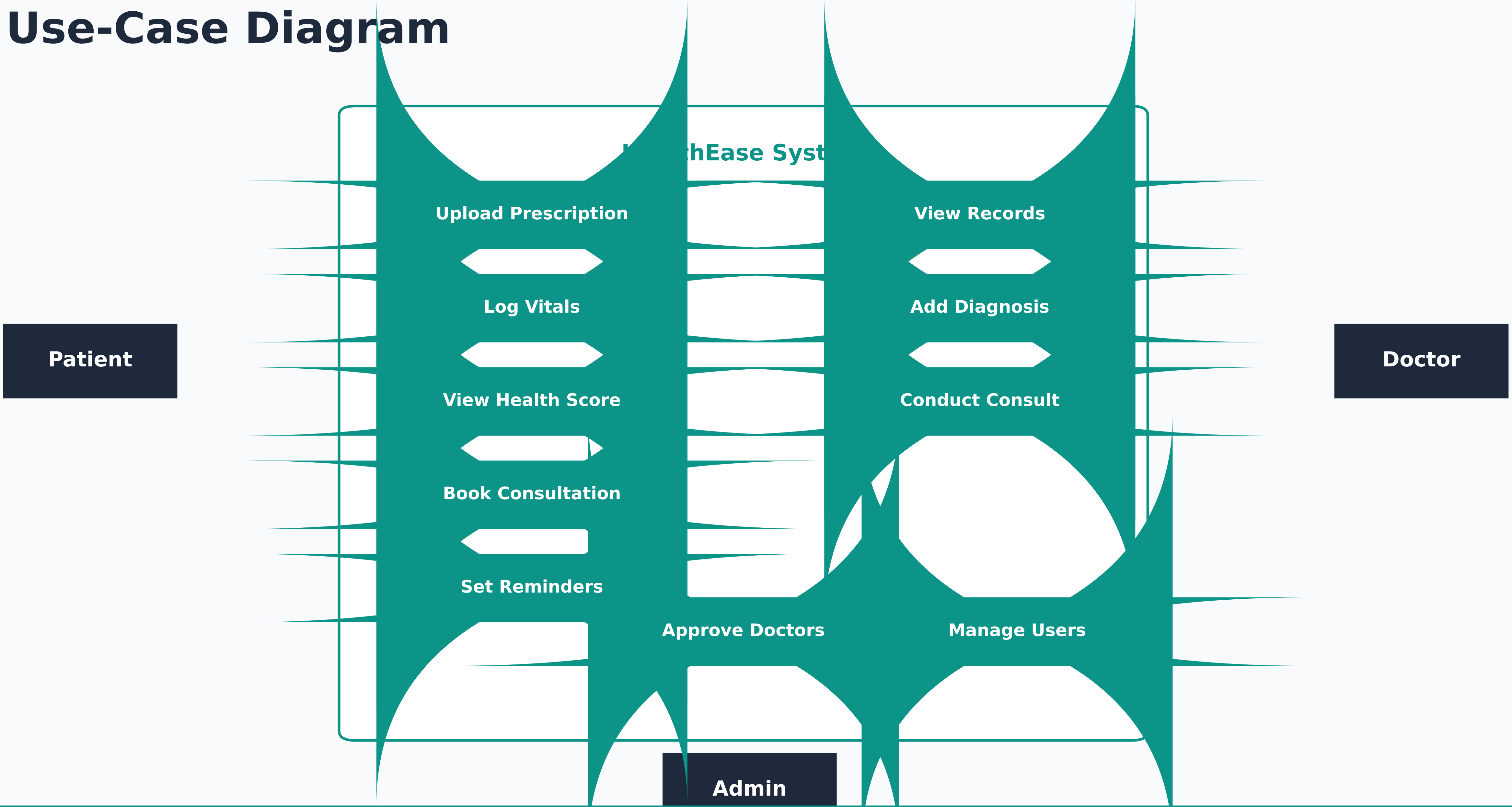
SYSTEM DESIGN

Data Flow Diagram (Level 1)



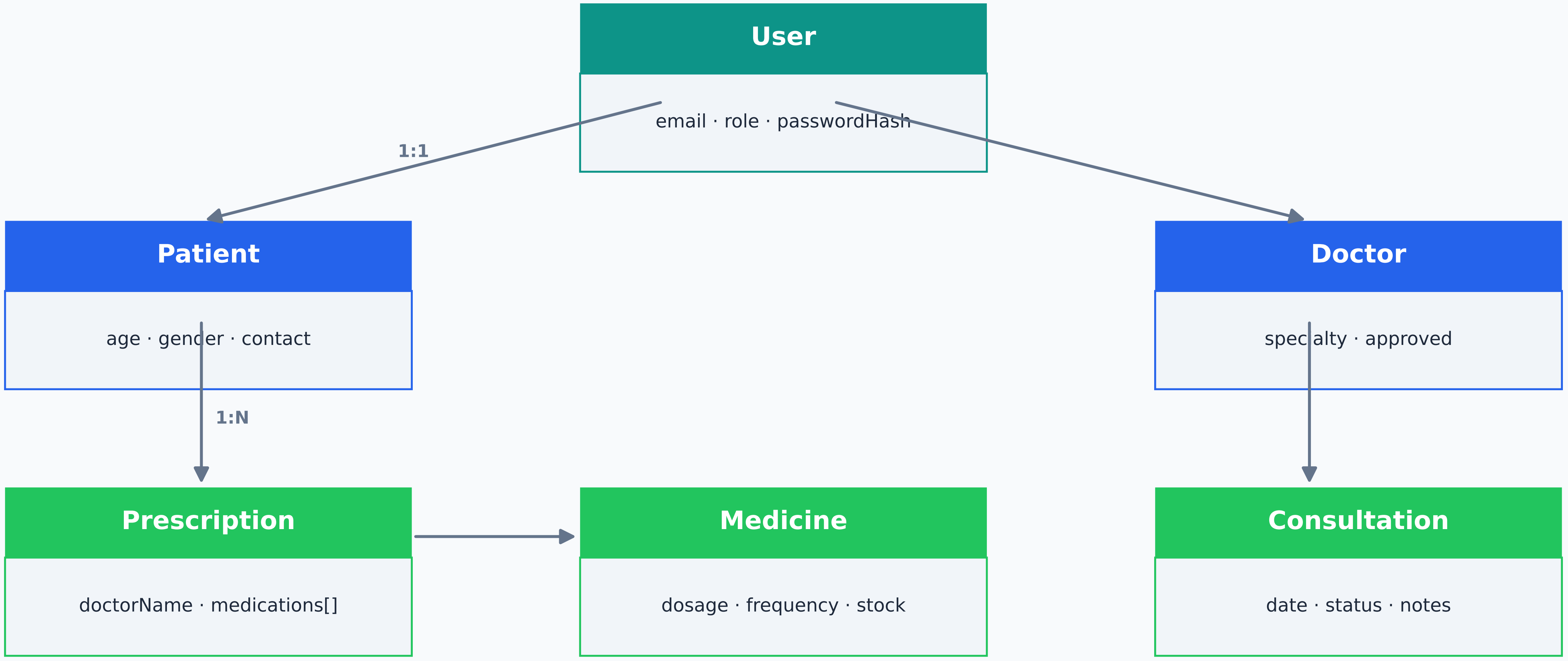
SYSTEM DESIGN

Use-Case Diagram



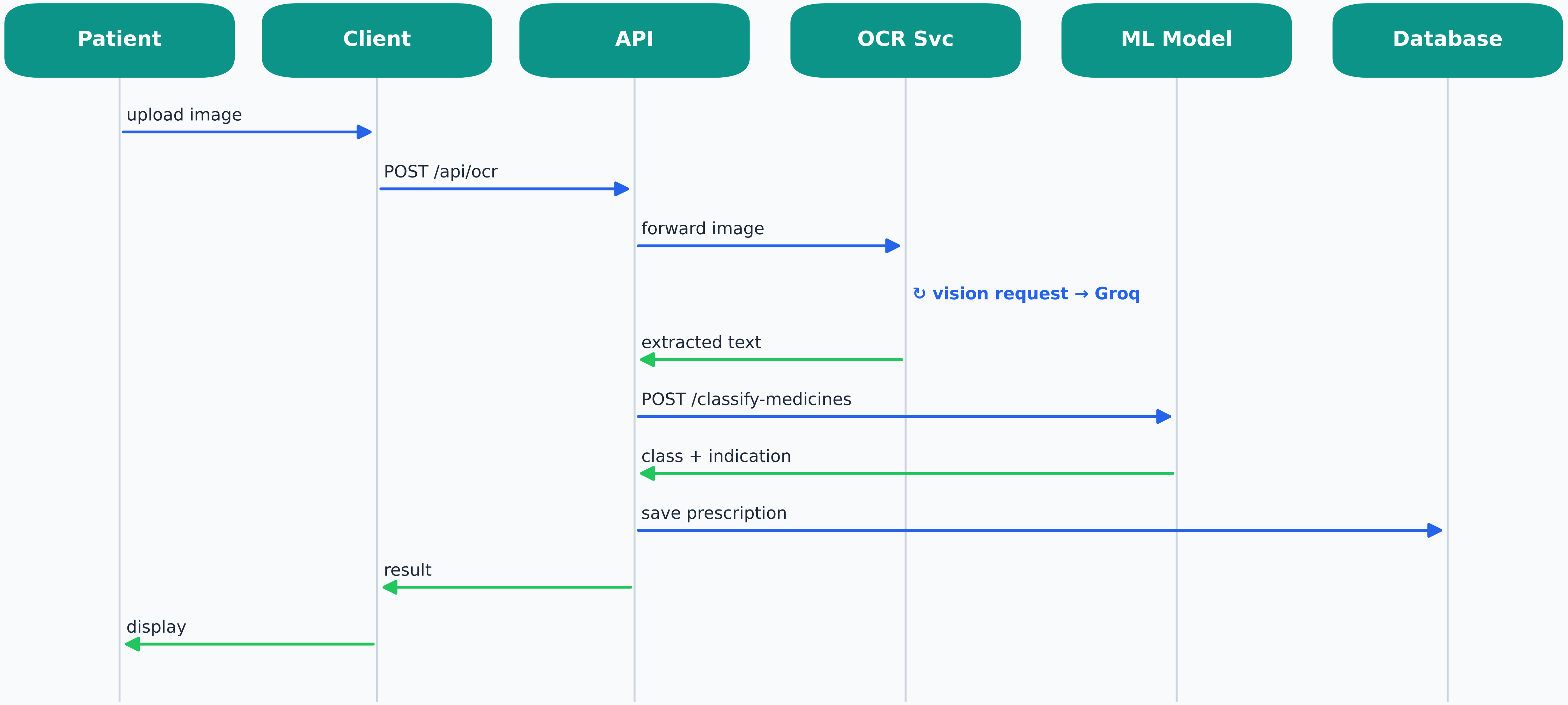
SYSTEM DESIGN

Database Design (ER Diagram)



SYSTEM DESIGN

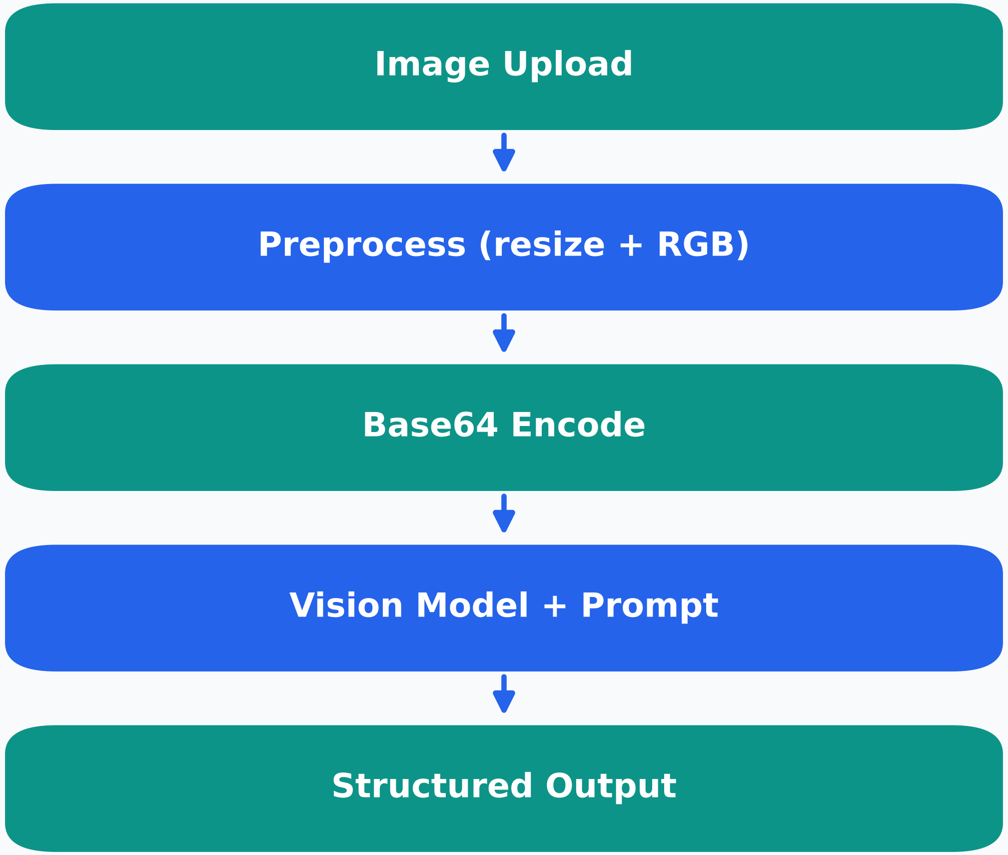
Sequence Diagram — Prescription Flow



IMPLEMENTATION

OCR Prescription Reader

- Type: AI Vision-Language OCR (not Tesseract).
- Engine: Groq Llama-4 Scout vision model (pre-trained).
- Handles printed AND handwritten prescriptions.
- Returns structured fields ready for the database.



IMPLEMENTATION · ML

Medicine-Indication Classifier

- Predicts therapeutic class + disease from the medicine NAME.
- Generalises to drug names never seen in training.
- Built on WHO INN naming stems (e.g. -pril, -statin).
- Custom-trained — reproducible from our dataset.

Stem	Class	Example
-pril	ACE Inhibitor	Lisinopril
-statin	Statin	Atorvastatin
-cillin	Penicillin	Amoxicillin
-floxacin	Fluoroquinolone	Ciprofloxacin
-prazole	PPI	Omeprazole
-dipine	Ca-Channel Blocker	Amlodipine

ML Methodology

Curated Dataset — 321 medicines · 27 classes

↓

Feature Extraction — Character n-gram TF-IDF (n = 2–4)

↓

Train / Test Split — 75% / 25% stratified

↓

Model Training — Logistic Regression

↓

5-Fold Cross-Validation

↓

Evaluation — Accuracy · F1 · Confusion Matrix

↓

Deployed Model — medicine_classifier.joblib

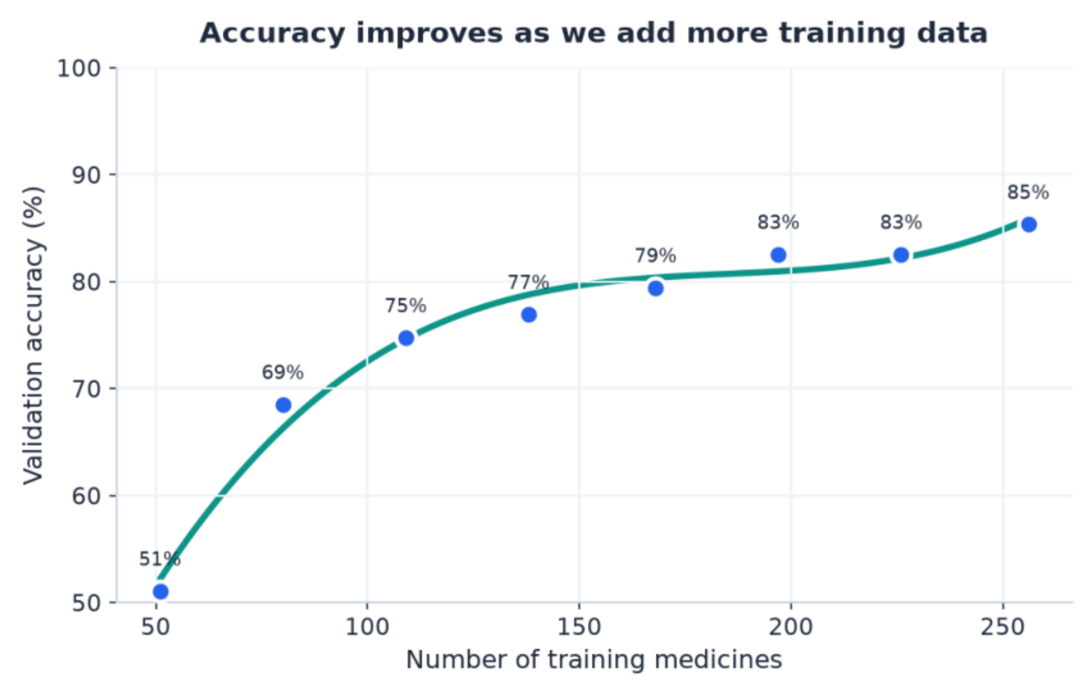
IMPLEMENTATION · ML

ML Results

85.4%	76.5%	0.78
Cross-Validated	Test Accuracy	Macro F1

Model comparison (5-fold CV):

Logistic Regression	82.9%
Linear SVM	81.7%
Naive Bayes	77.9%



The model independently rediscovered medical naming conventions from data.

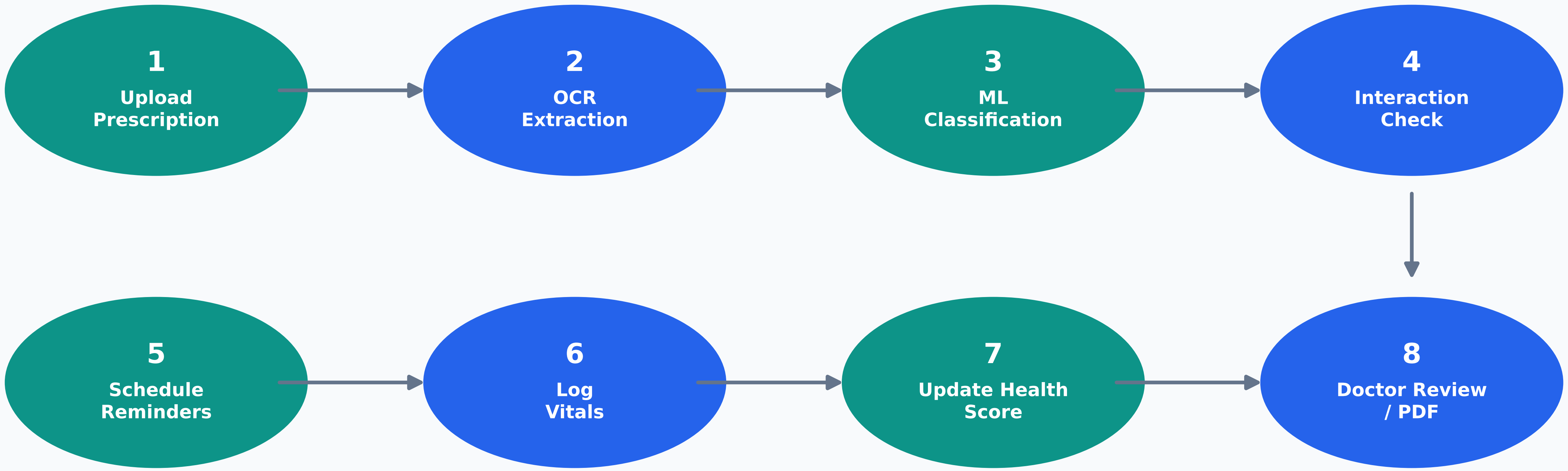
IMPLEMENTATION

Key Features

1. **Multi-role Auth**
2. **AI OCR Digitiser**
3. **Medicine→Disease**
4. **Interaction Check**
5. **Vitals Analytics**
6. **Smart Health Score**
7. **Tracker + Reminders**
8. **Notifications**
9. **Telemedicine**
10. **Admin Dashboard**
11. **PDF Export**
12. **Dark Mode / UI**

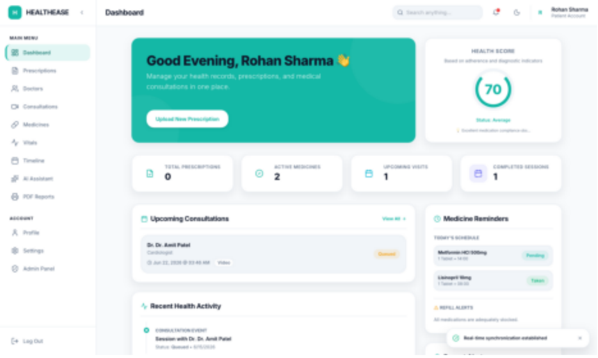
IMPLEMENTATION

End-to-End Workflow

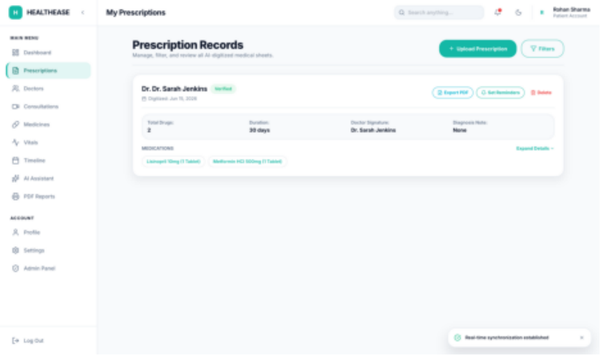


RESULTS & TESTING

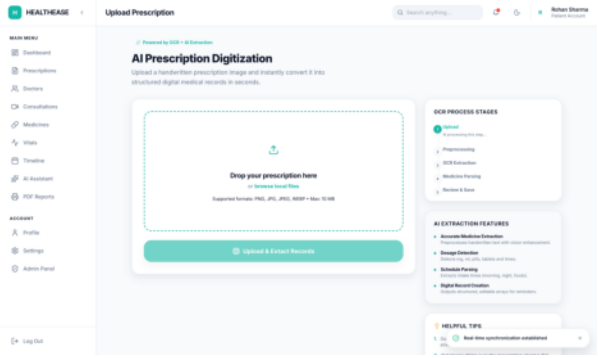
Application Results



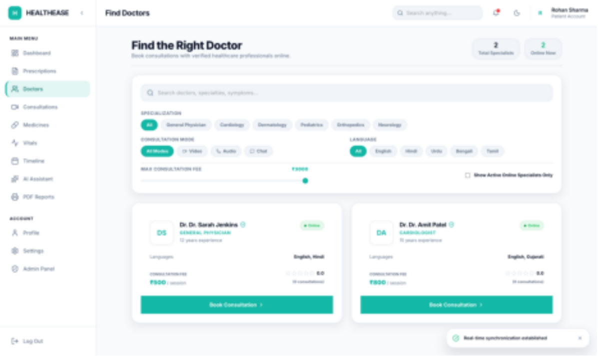
Patient Dashboard



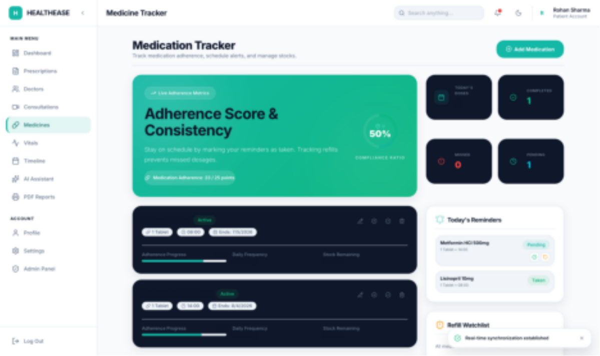
Prescription Records



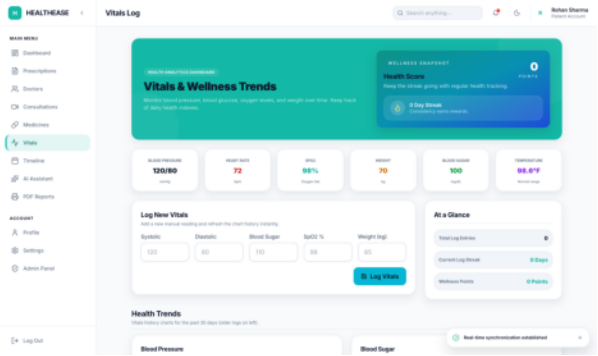
Prescription Digitizer



Doctor Marketplace



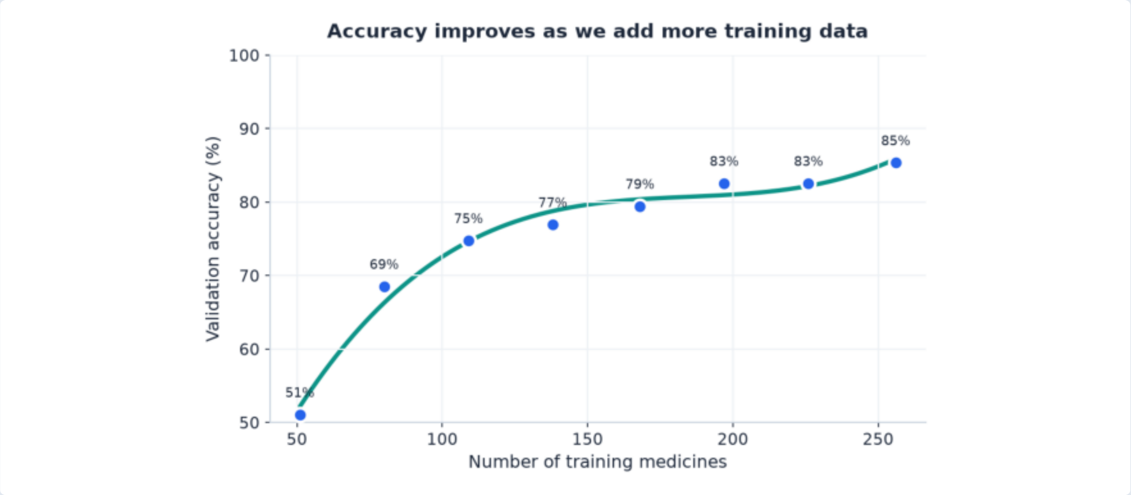
Medicine Tracker

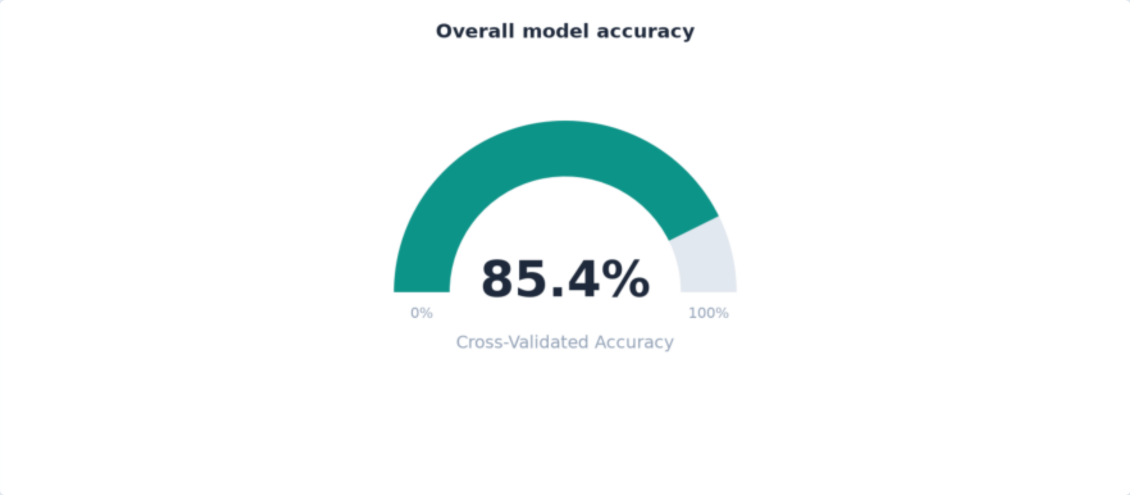


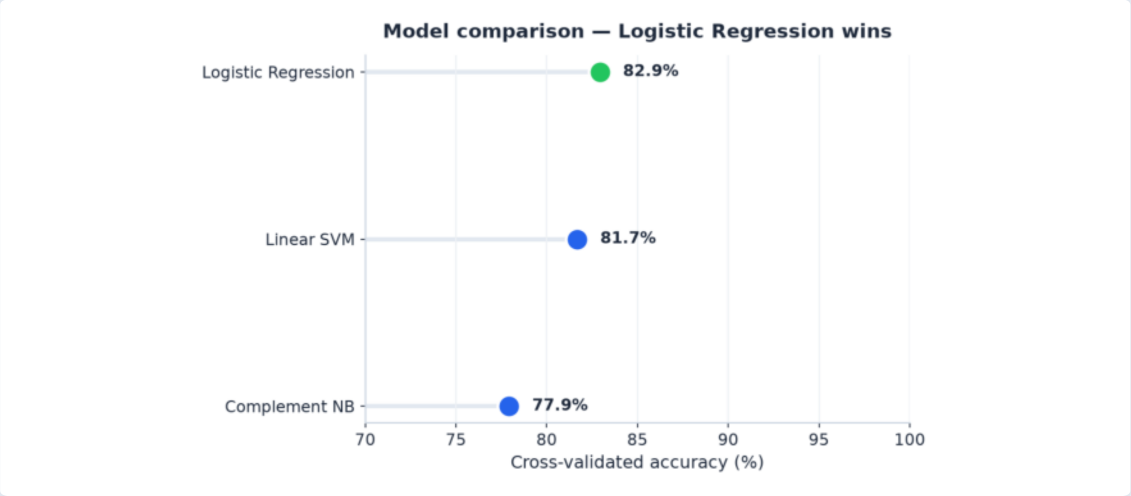
Vitals Analytics

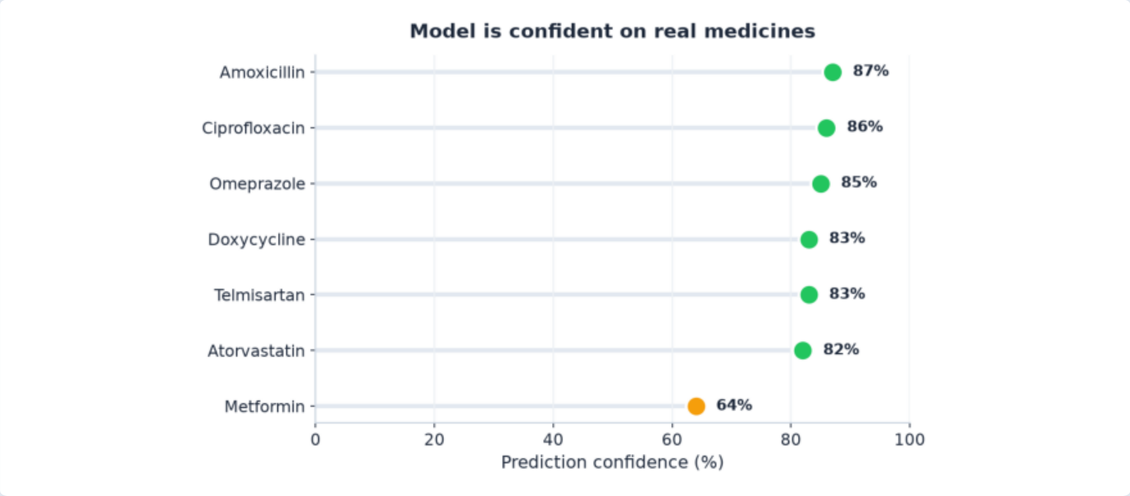
RESULTS & TESTING

ML Graphs & Analytics









RESULTS & TESTING

Testing

ID	Module	Input	Expected	Result
T01	Auth	Valid login	JWT returned	**Pass**
T02	Auth	Wrong password	401 error	**Pass**
T03	OCR	Prescription image	Structured fields	**Pass**
T04	ML Classify	"Amoxicillin"	Bacterial Infection	**Pass**
T05	Interactions	2 drugs	Interaction list	**Pass**
T06	Reminder	Set 08:00	Reminder saved	**Pass**
T07	Health Score	Vitals logged	Score 0–100	**Pass**

CONCLUSION

Future Scope

1. WebRTC live video consults
2. Wearable sync (Apple/Google Fit)
3. Predictive ML for vitals
4. Expand to 50+ drug classes
5. Mobile app

Thank You